MATH 4610: Fundamentals of Numerical Analysis Fall 2026

Final Project B

Due: Friday, December 18, 2026, 11:59 pm, on Canvas

General Instructions

This is one of three final projects . Choose one of them. Submit on Canvas (i) a report in PDF, typeset in LaTeX or Word, and (ii) a zip file with your code. All code must be written in MATLAB. Include one script for each problem (`Q1.m`, `Q2.m`, `Q3.m`) that produces the figures and tables of that problem, and a short `README` file. For each problem, the report should contain the derivations, the tables and figures, and a brief discussion of the results. Do not use MATLAB's built-in solvers such as `fzero`, `fsolve`, or `fminunc`, except to check your answers.

You may discuss the project with others, but your code and your report must be your own. Cite any sources you use.

Final presentation. Each student will present the project in class during the week of December 7–11, on Monday (Dec. 7), Wednesday (Dec. 9), or Friday (Dec. 11). The schedule will be posted on Canvas. Present the problem, your main results, and your conclusions; the work need not be finished. The written report is due on December 18 and may take into account comments made at the presentation.

Grading

Problem 1: roots of a scalar equation	20
Problem 2: a system of nonlinear equations	25
Problem 3: unconstrained minimization	25
Code (correct, readable, reproducible)	10
Written report	10
Final presentation	10
Total	100
Bonus: a constrained problem	10

Problem 1: Roots of a Scalar Equation (20 points)

The equation

$$f(x) = x^3 - 3x + 1 = 0$$

has three real roots. The substitution $x = 2\cos\theta$ shows that they are $2\cos(2\pi/9)$, $2\cos(4\pi/9)$, and $2\cos(8\pi/9)$; use these as exact values x^*.

(a) Write MATLAB functions for the bisection method, Newton's method, and the secant method. Stop when $|f(x_k)| < 10^{-12}$ or $|x_{k+1} - x_k| < 10^{-12}$, or after 100 iterations. Find all three roots with each method and make a table of the numbers of iterations. For bisection, compare the number of iterations with the bound $n \geq \log_2\big((b-a)/\mathrm{tol}\big)$.

(b) Let $e_k = |x_k - x^*|$. For one of the roots, plot e_k against k on a semilog scale for the three methods, and compute

$$p_k = \frac{\log(e_{k+1}/e_k)}{\log(e_k/e_{k-1})}.$$

Compare with the theoretical orders: 1 for bisection, 2 for Newton's method, and $(1+\sqrt{5})/2 \approx 1.618$ for the secant method.

(c) Run Newton's method from at least five starting points, some of them close to $x = \pm 1$, where $f'(x) = 0$. For each run, report which root is found and the number of iterations. Explain what happens near $x = \pm 1$.

(d) Apply Newton's method to $g(x) = x^3 - 3x + 2 = (x-1)^2(x+2)$ with $x_0 = 2$. Show numerically that $e_{k+1}/e_k \to 1/2$, so that the convergence is only linear, and explain this with a Taylor expansion of g about $x^* = 1$. Then use the modified iteration $x_{k+1} = x_k - 2g(x_k)/g'(x_k)$ and show that it converges quadratically.

Problem 2: A System of Nonlinear Equations (25 points)

Consider the system

$$x^2 + y^2 = 1, \qquad x^3 - y = 0,$$

whose solutions are the points where the unit circle meets the curve $y = x^3$. Write it as $F(\mathbf{x}) = 0$ with

$$F(x, y) = \begin{bmatrix} x^2 + y^2 - 1 \\ x^3 - y \end{bmatrix}.$$

(a) Find the Jacobian matrix $J(x, y)$ and the set of points where it is singular. Show that $t = x^2$ satisfies $t^3 + t - 1 = 0$ at a solution, and use this to show that the system has exactly two real solutions. Compute them to full accuracy; call them $\mathbf{x}^*$.

(b) Implement Newton's method

$$J(\mathbf{x}_k)\,\boldsymbol{\delta}_k = -F(\mathbf{x}_k), \qquad \mathbf{x}_{k+1} = \mathbf{x}_k + \boldsymbol{\delta}_k,$$

solving the linear system at each step (do not compute J^{-1}). Stop when $\|F(\mathbf{x}_k)\|_2 < 10^{-12}$ or after 50 iterations. For at least three starting points, plot $\|\mathbf{x}_k - \mathbf{x}^*\|_2$ against k, and plot the iterates (x_k, y_k) together with the two curves.

(c) Implement Broyden's method with $B_0 = J(\mathbf{x}_0)$ and

$$B_{k+1} = B_k + \frac{(\mathbf{y}_k - B_k \mathbf{s}_k)\,\mathbf{s}_k^T}{\mathbf{s}_k^T \mathbf{s}_k}, \qquad \mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k, \quad \mathbf{y}_k = F(\mathbf{x}_{k+1}) - F(\mathbf{x}_k).$$

Compare it with Newton's method: number of iterations, number of Jacobian evaluations, and the ratios $\|\mathbf{x}_{k+1} - \mathbf{x}^*\| / \|\mathbf{x}_k - \mathbf{x}^*\|$.

(d) Take a 201×201 grid of starting points in $[-2, 2] \times [-2, 2]$. Run Newton's method from each point and color the point according to the solution it reaches, with a third color if it does not converge in 50 iterations. Draw the singular set of J on the same figure and describe what you see.

Problem 3: Unconstrained Minimization (25 points)

Minimize the Rosenbrock function

$$f(x, y) = (1 - x)^2 + 100(y - x^2)^2,$$

whose only minimizer is $(1, 1)$. Start from $\mathbf{x}_0 = (-1.2,\ 1)$ and stop when $\|\nabla f(\mathbf{x}_k)\|_2 < 10^{-8}$. In every method, choose the step length by backtracking: start with $\alpha = 1$ and halve α until

$$f(\mathbf{x}_k + \alpha \mathbf{p}_k) \le f(\mathbf{x}_k) + 10^{-4}\, \alpha\, \nabla f(\mathbf{x}_k)^T \mathbf{p}_k.$$

(a) Compute ∇f and the Hessian matrix H, and check them at a few points with centered differences. Find the eigenvalues and the condition number of $H(1, 1)$.

(b) Implement the gradient (steepest descent) method, $\mathbf{p}_k = -\nabla f(\mathbf{x}_k)$. Stop after 5×10^4 iterations if the tolerance has not been reached.

(c) Implement Newton's method, $H(\mathbf{x}_k)\mathbf{p}_k = -\nabla f(\mathbf{x}_k)$. If $H(\mathbf{x}_k)$ is not positive definite (test this with `chol`), use either $\mathbf{p}_k = -\nabla f(\mathbf{x}_k)$ or the matrix $H(\mathbf{x}_k) + \mu I$ with $\mu > 0$ large enough; say which you use.

(d) Implement the BFGS method. Let $M_0 = I$, $\mathbf{p}_k = -M_k \nabla f(\mathbf{x}_k)$, and

$$M_{k+1} = (I - \rho_k \mathbf{s}_k \mathbf{y}_k^T)\, M_k\, (I - \rho_k \mathbf{y}_k \mathbf{s}_k^T) + \rho_k \mathbf{s}_k \mathbf{s}_k^T, \qquad \rho_k = \frac{1}{\mathbf{y}_k^T \mathbf{s}_k},$$

where $\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$ and $\mathbf{y}_k = \nabla f(\mathbf{x}_{k+1}) - \nabla f(\mathbf{x}_k)$. Keep $M_{k+1} = M_k$ if $\mathbf{y}_k^T \mathbf{s}_k \le 0$.

(e) For the three methods, plot $f(\mathbf{x}_k)$ and $\|\nabla f(\mathbf{x}_k)\|_2$ against k on a semilog scale, and make a table of the numbers of iterations, of function, gradient, and Hessian evaluations, and of the computing times. Draw the level curves of f (with logarithmically spaced levels) and the path of each method. Use the condition number found in (a) to explain the zig-zag path of the gradient method, and compare the observed rates of convergence.

Bonus: A Constrained Problem (10 points)

Minimize the Rosenbrock function subject to $g(x, y) = x + y - 1 = 0$. Write the Lagrange conditions

$$\nabla f(x, y) + \lambda \nabla g(x, y) = 0, \qquad g(x, y) = 0$$

as a system of three equations $G(x, y, \lambda) = 0$, find its Jacobian matrix, and solve the system with your Newton code from Problem 2. Check the answer by putting $y = 1 - x$ and minimizing the resulting function of one variable with your code from Problem 1 applied to its derivative.